

Random Graph Models: Theory, Simulation, and Modern Applications

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1 Introduction

1.1 Motivation and Background

When approaching a mathematical problem, a natural first instinct is to try and visualize the underlying idea. While this strategy is often effective in low-dimensional or simple settings, it quickly becomes impractical when dealing with high-dimensional spaces, large datasets, or complex networks. In such situations, intuition based on direct visualization tends to break down.

Graphs provide a powerful abstraction that overcomes this limitation. By explicitly representing relationships between discrete objects, graphs allow complex and seemingly overwhelming systems to be reduced to a collection of vertices and edges. In this way, an easy-to-work-with structure emerges from complexity: one may quite literally "connect the dots" and analyze the resulting network using mathematical tools.

In many real-world systems, however, these connections do not arise according to a fixed or deterministic pattern. Instead, interactions are often irregular, uncertain, or influenced by chance. Random graph models formalize this randomness by treating edges as probabilistic objects, thereby providing a framework in which typical structural properties, large-scale behavior, and rare but significant events can be understood, worked with and further studied.

1.2 Random Graphs in Theory and Practice

From a theoretical perspective, one moves away from "What does this represent in the real-world" and towards "what are the properties of this probability space?".

One of the most striking discoveries in random graph theory is the existence of sharp phase transitions [1]. In certain models, a small change in a single parameter can cause a sudden and dramatic shift in its structure, such as the sudden emergence of a giant connected component, an ubiquitous phenomenon.

In practice, random graphs play a crucial role as a baseline for real-world networks. By comparing an observed network with a suitable random graph, one can distinguish structural features that are of significance from those that are likely to arise by chance. This approach is widely used in fields such as network science, biology, economics, and computer science, where it is often infeasible to model every individual interaction deterministically (i.e., where precise inputs do not yield fixed certainties, but rather random, yet statistically foreseeable, outputs).

The classic Erdős–Rényi model is beloved for its simplicity. It describes a simple, egalitarian world where every connection is equally likely. This mathematical simplicity allows us to prove beautiful theorems about how networks evolve.

But the real world is not, unfortunately, as equal. Real networks, from the Internet to social media, are dominated by hubs, where the degree is huge, which are superstars that dominate the network. While the classic model predicts that everyone should have roughly the same number of friends, reality is full of massive outliers that define the system's behavior.

This gap between theory and reality forced an evolution. Modern random graph models have adapted to include this inequality. They retain the rigorous probabilistic framework of the past, but finally give us the flexibility to model the heavy-tailed (outliers), hub-dominated networks we actually see naturally occurring.

1.3 Structure of the Seminar Paper

This seminar paper is organized to gradually develop the theory of random graphs, beginning with classical models and progressing toward modern applications in complex network analysis. Each section builds on the previous ones, moving from foundational concepts to increasingly realistic models and practical use cases.

Section 2 introduces the classical random graph models $G(n, M)$ and $G(n, p)$. Their definitions, basic properties, and relationships are discussed, with particular emphasis on expected degrees and typical structural behavior. These models serve as a mathematical baseline for understanding randomness in networks.

In Section 3, we examine one of the central phenomena in random graph theory: the phase transition in $G(n, p)$. We describe the subcritical, critical, and supercritical regimes and explain the emergence of a giant connected component, supported by intuitive proof sketches.

Section 4 complements the theoretical analysis with simulations. By visualizing random graphs across different parameter regimes, we illustrate how global structure changes as edge probabilities vary and confirm the theoretical predictions numerically.

Section 5 introduces the Chung–Lu random graph model, motivated by the limitations of classical models. We present its definition, derive its expected degree properties, and explain how it enables the construction of graphs with prescribed degree sequences.

Building on this, Section 6 discusses scale-free networks and power-law degree distributions, highlighting their relevance for real-world systems and their structural implications. Section 7 then applies these models to the analysis of systemic risk and rare events, including contagion phenomena and the probabilistic detection of anomalous subgraphs.

Finally, Section 8 presents an advanced application of random graph theory to probabilistic link prediction. By interpreting an observed network as a realization of an underlying random graph model, we frame future edge formation as an inference problem grounded in probabilistic reasoning.

1.4 Graph-Theoretic Preliminaries and Notation

We briefly recall standard graph-theoretic notation used throughout the paper. A (simple, undirected) graph $G = (V, E)$ consists of a finite vertex set V and an edge set $E \subseteq \{\{i, j\} : i \neq j, i, j \in V\}$. The degree $d(v)$ of a vertex v is the number of edges incident to it. A connected component is a maximal subset of vertices connected by paths. We write $n = |V|$ and $m = |E|$.

In figure 1 we can see 3 components: a connected component (nodes 1 to 5) featuring a "supperstar" (node 4), with a much higher degree (4) than the rest. And two smaller components (nodes 6 and 7) and node 8, an isolated vertex.

A Breadth-First (later used in the paper) has the following idea: you start the search and visit a node, then you visit all of its neighbors, the other connected nodes. After this you pick the next node and do the same. After doing this you go back to the original list of nodes (the neighbors of the start node) and you pick the next node, and so on. Here we start from 1, we visit 4 and 5. Then we see that 1 has only one connection, through an already explored node. Then we go to 4 and we discover 2 and 3. If we had started from node 4 directly, the entire connected graph would have been explored.

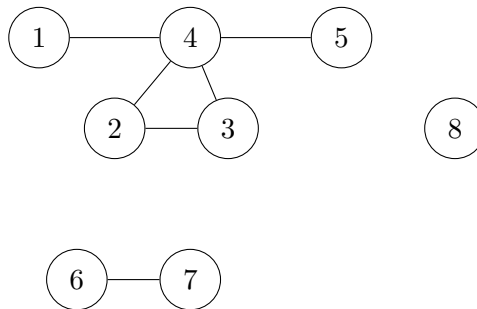


Figure 1: Example of a simple undirected graph illustrating basic graph-theoretic concepts. The graph consists of multiple connected components, including an isolated vertex.

2 Classical Random Graph Models

2.1 The Uniform Random Graph $G(n, M)$

The random graph model $G(n, M)$, originally studied by Erdős and Rényi (1959)[1], is the simplest model of a random graph with a fixed number of edges.

We begin with a fixed set of n vertices, $V = \{1, 2, \dots, n\}$. And let N be the total number of distinct pairs of vertices in V . This is the maximum possible number of edges a simple graph on n vertices can have (the maximum number of vertex pairs)[1] :

$$N = \binom{n}{2} = \frac{n(n-1)}{2}.$$

The sample space, denoted $\mathcal{G}_{n,M}$, consists of all graphs on V that have exactly M edges. The size of this sample space is determined by simple combinatorics: we choose a subset of size M (the edges present in the graph) from a set of size N (all possible edges). Thus, the total number of such graphs is given by the binomial coefficient $\binom{N}{M}$.

The model is called "uniform" because every graph $G \in \mathcal{G}_{n,M}$ is equally likely to be picked. Since there are $\binom{N}{M}$ total graphs, the probability of observing any single specific graph G is given by [1]:

$$\mathbb{P}(G) = \frac{1}{|\mathcal{G}_{n,M}|} = \binom{N}{M}^{-1}.$$

Consequently, if a graph has a number of edges different from M , its probability is zero.

Although conceptually straight-forward, $G(n, M)$ is often difficult to analyze due to the lack of independence between edges. Knowing that an edge exists between two vertices slightly decreases the probability of an edge existing elsewhere, since the sample size for edges gets smaller. In order to deal with this issue, we will take a look at the binomial random graph.

2.2 The Binomial Random Graph $G(n, p)$

The binomial random graph model, denoted $G(n, p)$, provides an alternative formulation that helps us go around some problems found in $G(n, M)$ by introducing independence between edges.

Definition and Probability Space

As with the uniform model, we begin with a vertex set $V = \{1, \dots, n\}$ and the set of all $N = \binom{n}{2}$ possible pairs of vertices. However, instead of fixing the total number of edges, we determine the presence of each edge through a random process.

For every possible pair of distinct vertices $\{i, j\}$, we perform an independent Bernoulli trial (a coin flip). We include the edge $\{i, j\}$ in the graph with probability p and exclude it with probability $1 - p$. The parameter $0 \leq p \leq 1$ is fixed and applies uniformly to all trials. [1]

Probability of a Specific Graph

Because the edges are selected independently, the probability of observing a specific graph G depends only on its number of edges, denoted by $m = |E|$.

To obtain a specific graph G with exactly m edges, two conditions must be met simultaneously:

1. The m specific pairs in G must be chosen (each with probability p).
2. The remaining $N - m$ pairs must *not* be chosen (each with probability $1 - p$).

Since these choices are independent events, we multiply their probabilities to find the probability of the graph G , thus obtaining the :

$$\mathbb{P}(G) = p^m(1 - p)^{N-m}.$$

Unlike in $G(n, M)$, the total number of edges in $G(n, p)$ is not fixed. Instead, the number of edges is a random variable that follows a binomial distribution with parameters N and p . This distribution gives the model its name: the probability that a graph in $G(n, p)$ has exactly k edges is:

$$\mathbb{P}(|E| = k) = \binom{N}{k} p^k (1 - p)^{N-k}.$$

Thus, $G(n, p)$ can be viewed as a version of $G(n, M)$ where the number of edges fluctuates around an expected value of $\mathbb{E}[|E|] = pN$.

2.3 Expected Degree and Basic Properties

Before jumping into formal mathematics we should first get an understanding of what "expected degree" actually is.

Informal Understanding

Consider a social gathering with n people, where every person is a vertex -v. Your degree is simply the number of friends you have.

In the **Binomial Random Graph** $G(n, p)$, making friends is analogous to flipping a biased coin. For every other person in the room ($n - 1$ people), you flip a coin that lands on "Heads" (friendship) with probability p . Since you repeat this experiment $n - 1$ times independently, if your success rate is p , you naturally expect to succeed $(n - 1)p$ times.

In the **Uniform Random Graph** $G(n, M)$, the mechanism is different. Imagine a party with n guests where the host creates a strict rule: there can be exactly M friendships in the entire room. To visualize this, imagine the host prints out M "Friendship Contracts". These contracts are then assigned to M pairs of guests chosen completely at random from all possible pairs. This creates a subtle dependency: if Akshay and Anna are assigned a contract, that is one less contract available for everyone else. However, the system remains perfectly "fair" because every pair has the same probability of being chosen ($1/n$, probability for all, we pick one, $1/n - 1$, for all). Since each of the M contracts is between two people (two signatures), there are $2M$ "spots" for friends available in total. When these spots are shared among the n guests, the "fair share"—or the average number of friends per person is $2M/n$.

Formal Derivation of Expected Degree

We will now rigorously prove this.

Indicator Let v be an arbitrary vertex in V . We now want to compute the expected degree $\mathbb{E}[\deg(v)]$. We define an indicator of a random variable 1_{vu} for every other vertex $u \in V \setminus \{v\}$:

$$1_{vu} = \begin{cases} 1 & \text{if the edge } \{v, u\} \text{ exists,} \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

The degree of vertex v is simply the sum of these indicators over all possible neighbors:

$$\deg(v) = \sum_{u \neq v} 1_{vu}.$$

We now use the fact that the expectation is linear, which means that the expected value of a sum is the sum of the expected values, $\mathbb{E}[\sum Y_i] = \sum \mathbb{E}[Y_i]$.

$$\mathbb{E}[\deg(v)] = \mathbb{E}\left[\sum_{u \neq v} 1_{vu}\right] = \sum_{u \neq v} \mathbb{E}[1_{vu}].$$

Now we evaluate this sum for each model.

The Binomial Model $G(n, p)$ In $G(n, p)$, every edge exists independently with probability p . The expected value of an indicator variable is simply the probability that it equals 1:

$$\mathbb{E}[1_{vu}] = 1 \cdot \mathbb{P}(\{v, u\} \in E) + 0 \cdot \mathbb{P}(\{v, u\} \notin E) = p.$$

Substituting this back into our sum, which has exactly $n - 1$ terms (one for each potential neighbor): [2]

$$\mathbb{E}[\deg(v)] = \sum_{u \neq v} p = (n - 1)p.$$

[2]

Case 2: The Uniform Model $G(n, M)$ Let $N = \binom{n}{2}$ be the total number of possible pairs of vertices. In this model, we select exactly M edges uniformly at random from these N pairs. The probability that any specific pair $\{v, u\}$ is selected is the ratio between the graphs that have this very specific edge and the total number of graphs in $G_{n, M}$.

$$\mathbb{P}(\{v, u\} \in E) = \frac{\binom{N-1}{M-1}}{\binom{N}{M}} = \frac{M}{N}.$$

Using the same linearity of expectation argument as above:

$$\mathbb{E}[\deg(v)] = \sum_{u \neq v} \mathbb{E}[1_{vu}] = (n - 1) \frac{M}{N}.$$

Substituting $N = \frac{n(n-1)}{2}$, we get:

$$\mathbb{E}[\deg(v)] = (n - 1) \frac{M}{\frac{n(n-1)}{2}} = (n - 1) \frac{2M}{n(n - 1)} = \frac{2M}{n}.$$

[2]

Further Structural Properties

While the expected degree provides an overall, centralized value, we still don't have a clear enough image in our heads. How are the degrees distributed around this mean? How fast do we actually get to an outlier (e.g. a "superstar"), how likely are we to have a triangle (i.e. 3 nodes, isolated from the rest of the graph and each of them has a degree of 2).

1. **Degree Distribution (Poisson Limit):** In the sparse regime where $n \rightarrow \infty$ and the average degree is constant ($p = \lambda/n$), the degree distribution of $G(n, p)$ converges to a Poisson distribution.

Proof: We start with the exact probability that a vertex v has degree k , which follows a Binomial distribution $B(n - 1, p)$:

$$\mathbb{P}(\deg(v) = k) = \binom{n - 1}{k} p^k (1 - p)^{n - 1 - k}.$$

We substitute $p = \frac{\lambda}{n}$. As $n \rightarrow \infty$, we can break this expression down step-by-step:

$$\begin{aligned}
\mathbb{P}(\deg(v) = k) &= \frac{(n-1)(n-2)\dots(n-k)}{k!} \cdot \left(\frac{\lambda}{n}\right)^k \cdot \left(1 - \frac{\lambda}{n}\right)^{n-1-k} \\
&\approx \frac{n^k}{k!} \cdot \frac{\lambda^k}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(1 - \frac{\lambda}{n}\right)^{-1-k} \\
&= \frac{1}{k!} \underbrace{\frac{n^k}{n^k}}_{\rightarrow 1} \lambda^k \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\rightarrow e^{-\lambda}} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-1-k}}_{\rightarrow 1} \\
&\rightarrow \frac{\lambda^k}{k!} e^{-\lambda}.
\end{aligned}$$

2. **Concentration of Measure:** Degrees in random graphs are tightly concentrated around the mean. Large deviations from the expected degree are exponentially rare. Unlike social networks, where we see a great variance in popularity.

Proof Sketch: The degree of a vertex v can be written as a sum of $n-1$ independent indicators, as used previously. If v is connected to u . Thus, $\deg(v)$ follows a Binomial distribution $\text{Bin}(n-1, p)$. We apply the **Chernoff Bound** [1], which states that for any $0 \leq \epsilon \leq 1$:

$$\mathbb{P}(|\deg(v) - \mu| \geq \epsilon\mu) \leq 2e^{-\frac{\epsilon^2\mu}{3}}.$$

This inequality shows that the probability of a vertex's degree deviating from the mean by a factor of ϵ decays exponentially with the expected degree. Therefore, for large n , the probability of finding significant outliers becomes almost insignificant. [1]

3. **Slow Growth of Maximum Degree:** The maximum degree $\Delta(G)$ grows very slowly. It can be shown that for sparse graphs [1],

$$\Delta(G) \approx \frac{\log n}{\log \log n}$$

almost surely. This limit is far lower than the maximum degrees observed in empirical networks like the internet.

Proof Sketch: We seek the smallest degree k such that it is unlikely for any vertex to have degree k . This occurs when the expected number of such vertices drops below 1:

$$n \cdot \mathbb{P}(\deg(v) = k) \approx 1.$$

Using the Poisson approximation $\mathbb{P}(\deg(v) = k) \approx \frac{\lambda^k}{k!} e^{-\lambda}$ and Stirling's approximation $k! \approx (k/e)^k$, we solve for k :

$$n \left(\frac{e\lambda}{k}\right)^k \approx 1 \quad \Rightarrow \quad \ln n \approx k \ln k.$$

This threshold remains very low even for massive n [1] [2].

4. **Low Clustering Coefficient:** The local clustering coefficient $C(v)$ measures the probability that two friends of v are also friends with each other (forming a triangle). [1]

Proof: In $G(n, p)$, edges are independent. If vertex v has neighbors u and w , the probability that the edge $\{u, w\}$ exists is simply p , regardless of the fact that they are both connected to v .

$$C(v) = \frac{\text{Actual Edges between Neighbors}}{\text{Possible Edges between Neighbors}} \approx p.$$

For sparse graphs where $p = \lambda/n$, $C(v) \rightarrow 0$ as $n \rightarrow \infty$. This contrasts sharply with real-world social networks, which exhibit high clustering.

Remark: Understanding the Chernoff Bound

To formally justify why the outliers are so rare, we use on the Chernoff Bound [1]. The derivation is as it follows:

Let $X = \sum X_i$ be the sum of independent Bernoulli trials (the edges in our graph) with mean μ . We wish to bound the upper tail $\mathbb{P}(X \geq (1 + \epsilon)\mu)$. Standard manipulations (exponentiating both sides and applying Markov's inequality) yield the generic bound for any $t > 0$:

$$\mathbb{P}(X \geq (1 + \epsilon)\mu) = \mathbb{P}(e^{tX} \geq e^{t(1+\epsilon)\mu}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{t(1+\epsilon)\mu}}.$$

Using the independence of X_i and the inequality $1 + x \leq e^x$, we bound the numerator:

$$\mathbb{E}[e^{tX}] = \prod \mathbb{E}[e^{tX_i}] = \prod (1 - p + pe^t) = \prod (1 + p(e^t - 1)) \leq \prod e^{p(e^t - 1)} = e^{\mu(e^t - 1)}.$$

Substituting this back, we obtain a function of t that we will minimize in order to find the supremum:

$$\mathbb{P}(X \geq (1 + \epsilon)\mu) \leq \exp \left(\mu(e^t - 1) - t(1 + \epsilon)\mu \right).$$

Finding the bound To find the optimal t , we will minimize the exponent $f(t) = e^t - 1 - t(1 + \epsilon)$. In doing so we will do the standard procedure of differentiating with respect to t and setting the result to zero:

$$f'(t) = e^t - (1 + \epsilon) = 0 \quad \Rightarrow \quad t = \ln(1 + \epsilon).$$

Substituting this optimal t back into our bound:

$$\mathbb{P}(X \geq (1 + \epsilon)\mu) \leq \left(\frac{e^\epsilon}{(1 + \epsilon)^{(1+\epsilon)}} \right)^\mu.$$

While this form is exact, it doesn't really say much to us, a more numerical and shorter value would be a lot better. We simplify it by analyzing the term in the parentheses and applying a logarithm.

$$\ln \left(\frac{e^\epsilon}{(1 + \epsilon)^{(1+\epsilon)}} \right) = \ln(e^\epsilon) - \ln((1 + \epsilon)^{(1+\epsilon)}) = \epsilon - (1 + \epsilon) \ln(1 + \epsilon).$$

We substitute the Taylor approximation $(1 + \epsilon) \ln(1 + \epsilon) \approx \epsilon + \frac{\epsilon^2}{2} - \frac{\epsilon^3}{6}$:

$$\epsilon - \left(\epsilon + \frac{\epsilon^2}{2} - \frac{\epsilon^3}{6} \right) = -\frac{\epsilon^2}{2} + \frac{\epsilon^3}{6}.$$

We require a clean upper bound. For $0 \leq \epsilon \leq 1$, we observe that $-\frac{\epsilon^2}{2} + \frac{\epsilon^3}{6} \leq -\frac{\epsilon^2}{3}$. Thus, exponentiating back, we obtain the standard form:

$$\left(e^{-\frac{\epsilon^2}{3}}\right)^\mu = e^{-\frac{\epsilon^2 \mu}{3}}.$$

An entirely analogous argument applied to the lower tail yields

$$\mathbb{P}(X \leq (1 - \epsilon)\mu) \leq e^{-\frac{\epsilon^2 \mu}{3}}.$$

Combining the upper and lower tail bounds using the union bound, we obtain

$$\mathbb{P}(|X - \mu| \geq \epsilon\mu) \leq \mathbb{P}(X \geq (1 + \epsilon)\mu) + \mathbb{P}(X \leq (1 - \epsilon)\mu) \leq 2e^{-\frac{\epsilon^2 \mu}{3}},$$

which is exactly the Chernoff bound used in the proof sketch.

2.4 Relationship Between $G(n, M)$ and $G(n, p)$

Having defined both the uniform model $G(n, M)$ and the binomial model $G(n, p)$, we could ask ourselves, how are they related? How can we compare them?

Intuitive Relationship

Intuitively, $G(n, p)$ can be viewed as a "freer" version of $G(n, M)$.

- In $G(n, M)$, the number of edges is a hard constraint: there are exactly M edges, no more, no less.
- In $G(n, p)$, the number of edges is a random variable. While the *expected* number of edges is Np , the actual number fluctuates slightly.

However, as the graph grows ($n \rightarrow \infty$), these fluctuations become negligible in comparison to the total size of the graph. A typical sample from $G(n, p)$ will have very close to $M \approx Np$ edges, almost resembling $G(n, M)$.

Mathematical Equivalence

The key mathematical link is that the Binomial model is simply a weighted average of Uniform models.

The Conditioning Property

If we take a random graph $G \sim G(n, p)$ and condition on the event that it happens to have exactly M edges, the resulting distribution is *exactly* $G(n, M)$.

$$\mathbb{P}_{G(n,p)}(G \mid |E(G)| = M) = \mathbb{P}_{G(n,M)}(G).$$

Proof: Let G_0 be a specific graph with exactly M edges. In the $G(n, p)$ model, the probability of observing this specific graph is $p^M(1-p)^{N-M}$. The probability of observing *any* graph with M edges is given by the binomial distribution: $\mathbb{P}(|E| = M) = \binom{N}{M} p^M (1-p)^{N-M}$. Applying the definition of conditional probability:

$$\mathbb{P}(G = G_0 \mid |E| = M) = \frac{\mathbb{P}(G = G_0)}{\mathbb{P}(|E| = M)} = \frac{p^M(1-p)^{N-M}}{\binom{N}{M} p^M (1-p)^{N-M}} = \frac{1}{\binom{N}{M}}.$$

This result, $\binom{N}{M}^{-1}$, is exactly the definition of the uniform model $G(n, M)$. Thus, $G(n, p)$ is essentially a weighted mixture of $G(n, M)$ spaces.

This means $G(n, p)$ is essentially a mixture of $G(n, M)$ graphs, weighted by the probability of having M edges.

Concentration of Edges

In $G(n, p)$, the number of edges $|E|$ follows a Binomial distribution $\text{Bin}(N, p)$. The previously mentioned and proved properties tell us that $|E|$ is extremely concentrated around its mean Np . The standard deviation is only $\sqrt{Np(1-p)}$, which is much smaller than the mean for large N . Thus, $G(n, p)$ spends almost all its probability mass on graphs that have approximately Np edges.[1]

3. Equivalence for Monotone Properties

This leads to a powerful equivalence theorem. Let $\mathcal{Q}$ be a **monotone property** (a property that is preserved if we add edges, such as "being connected" or "containing a triangle"). If we choose M and p such that $M \approx Np$, then as $n \rightarrow \infty$:

$$\mathbb{P}(G(n, M) \in \mathcal{Q}) \rightarrow \mathbb{P}(G(n, p) \in \mathcal{Q}).$$

The proof here is too long for the scope of this Seminar Paper, for it see Frieze & Karoński, Section 1.4 [1]

Application: The Transfer of Proofs

Why does this matter?

- **Ease of Work:** The independence of edges in $G(n, p)$ makes it mathematically much easier to work with. Calculating probabilities of subgraphs or degrees is straightforward multiplication.
- **Transfer Principle:** In many proofs (such as the Phase Transition or Connectivity), we perform the difficult calculations in the friendly setting of $G(n, p)$. Once the result is proven there, we use the equivalence theorems to assert that the same result holds for $G(n, M)$.

We will see this in action in the next section on Phase Transitions: the giant component emerges in $G(n, p)$ when $p = 1/n$, which corresponds exactly to the point $M = n/2$ in the uniform model.

3 The Phase Transition in $G(n, p)$

In our previous section we have covered what happens when we have an arbitrary p and an n that varies. Either low or tending to infinity, as seen in the convergence proofs. However, we have not yet played around with p , therefore we will do it now. We will split the p values in 3 regions, 2 regimes and a phase transition, one of the most fascinating parts of this paper.

3.1 The Subcritical Regime ($np < 1$)

When the expected degree is $np < 1$, the network lacks the density required for a long path to form. The graph is essentially a collection of unconnected dots.

Structure: A Forest of Trees

In this regime, the graph consists almost entirely of isolated trees. Cycles are extremely rare, as we have proved earlier that the clustering coefficient is anyways, low. The graph has no complex structure.

Component Size

The defining characteristic of this phase is that all connected components are microscopic relative to the total size of the graph. However, a valid question stays. How big could the largest connected component be?

The derivation is beyond the scope of this paper. However the result and the intuitive way of understanding it still interests us.

In the subcritical Erdős–Rényi random graph with edge probability $p = c/n$ for $c < 1$, it is shown that the size of the largest connected component grows at most logarithmically with n . More precisely, with high probability,

$$|C_{\max}| = \Theta(\log n).[1]$$

This result follows from bounding the expected number of connected components of a given size k . Consequently, the largest connected component is of order $\log n$.

3.2 The Critical Threshold ($np = 1$)

The case where $np = 1$ represents a transition point in the evolution of the Erdős–Rényi random graph. At this critical threshold, the graph undergoes a significant change in its global structure. While the graph is still sparse, it is no longer dominated purely by small tree components, nor has a giant connected component emerged.

Structure: Emergence of Large Components

The graph contains connected components of significantly larger size than in the subcritical regime. Although most components remain small, there is now a non-negligible probability of observing components whose size grows polynomially with n . These components are no longer necessarily just trees, cycles and more complex structures begin to appear.

Despite this growth, no single component captures a linear fraction of the vertices. The graph is therefore highly fragmented, but with occasional large clusters.

Component Size

At the critical threshold, the size of the largest connected component scales as a power of n . More precisely, with high probability,

$$|C_{\max}| = \Theta(n^{2/3}).[1]$$

This scaling represents a sharp increase compared to the logarithmic component sizes in the subcritical regime, yet it is still negligible relative to the total size of the graph.

3.3 The Supercritical Regime ($np > 1$)

When the expected degree exceeds one, the Erdős–Rényi random graph enters the supercritical regime. In this phase, the graph becomes dense enough to witness a high connectivity across all its nodes.

Structure: Emergence of a Giant Component

The defining structural feature of the supercritical regime is the appearance of a *giant connected component*. This component contains a significant chunk of all vertices in the graph, while the remaining vertices form only small, disconnected components

Component Size

In the supercritical phase, the Erdős–Rényi random graph with $p = c/n$ and $c > 1$, its component structure changes fundamentally. With high probability, the graph contains a unique connected component whose size is linear in n , while all other components remain comparatively small. In particular,

$$|C_{\max}| = \Theta(n).[1]$$

All remaining connected components do not participate in this structure. After removing the giant component, the residual graph behaves similarly to a subcritical random graph, and thus all other components are small. In particular, the size of the second-largest connected component is at most logarithmic in n with high probability. This establishes a clear separation between the giant component and the rest of the graph in the supercritical regime.

3.4 Proof Sketch via Branching Processes

Intuitive Idea

To analyze the size of connected components in $G(n, p)$, we have to model how a component grows when being explored, starting from a single vertex. The key idea here is that, for sparse graphs with $p = c/n$, the early stages of this exploration resemble a branching process.

The reason is that in the beginning, only a few vertices have been discovered, so edges leading back to the already explored vertices are not that common, something sustained by the previously mentioned clustering coefficient. Thus, each newly discovered vertex generates further, neighbours almost independently.

Exploration Process

We analyze the breadth-first exploration process of a component as described using the states *active*, *neutral*, and *inactive*. Starting with $S_0 = 1$ active vertex, after $t \geq 1$ explorations the number of active vertices satisfies the recursion

$$S_t = S_{t-1} + X_t - 1,$$

where X_t is the number of *neutral* vertices that become active when exploring the t -th active vertex. This recursion and its interpretation are part of the standard exploration process for $ER_n(p)$ (breadth-first order) [1]

In the Erdős–Rényi model $ER_n(p)$, conditionally on S_{t-1} , the random variable X_t is binomially distributed as

$$X_t | S_{t-1} \sim \text{Bin}(n - (t-1) - S_{t-1}, p),$$

because after $t-1$ explorations there are $t-1$ inactive vertices and S_{t-1} active vertices, leaving $n - (t-1) - S_{t-1}$ neutral vertices; each of them becomes active with probability p [1]

Now consider the sparse regime $p = \lambda/n$ (equivalently $p = c/n$). As long as the exploration is in its early stage (in particular, while t and S_{t-1} are not too large), the number of neutral vertices is still close to n , and thus X_t is close to a binomial $\text{Bin}(n, \lambda/n)$. Since $\text{Bin}(n, \lambda/n)$ converges to a Poisson random variable with mean λ , the early exploration history of a finite component in $ER_n(\lambda/n)$ is closely related to that of a Poisson branching process with parameter λ [1]. This is precisely the informal link developed in the notes, where finite exploration histories are shown to have asymptotically the same probabilities as in the Poisson branching setting [1]

The phase transition is reflected in the behavior of the largest component in the subcritical versus supercritical regimes. In the subcritical case $\lambda = np < 1$, the notes prove that the largest connected component is logarithmic: for every $a > 1/I_\lambda$,

$$\mathbb{P}_\lambda(|C_{\max}| \geq a \log n) = O(n^{-\delta}),$$

for some $\delta > 0$ (Theorem 4.4) [1]. This formalizes the statement that when $\lambda < 1$, all components remain small (of order $\log n$) with high probability.

For $\lambda > 1$ (supercritical regime), the notes state a law of large numbers for the size of the maximal component and explicitly identify the survival probability of the associated Poisson branching process as the quantity governing the giant component size (Theorem 4.8) :contentReference[oaicite:6]index=6. This establishes that the qualitative change in component sizes occurs at the threshold $\lambda = 1$, i.e. $np = 1$ (equivalently $p = 1/n$).

3.5 The Emergence of the Giant Component

Uniqueness of the Giant Component

From the exploration process, we have established that in the supercritical regime $p = c/n$ with $c > 1$, the exploration starting from a uniformly chosen vertex either terminates quickly, producing a component of logarithmic size, or grows to a larger size. This already suggests the existence of large connected components. We now show that such a component must be unique.

Let $C_{\max}$ denote the largest connected component of $G(n, p)$. Suppose that there exist two disjoint connected components, C_1 and C_2 , each of linear size.

$$|C_1| \geq \varepsilon n, \quad |C_2| \geq \varepsilon n$$

for some fixed $\varepsilon > 0$. We will now find a contradiction.

Consider the graph obtained after removing the vertices of C_1 . Since edges in $G(n, p)$ are independent, the induced subgraph on the remaining vertex set is again an Erdős–Rényi random graph with the same edge probability $p = c/n$, but on

$$n' = n - |C_1| = (1 - \varepsilon)n$$

vertices. The expected degree of this remaining graph is therefore

$$n'p = (1 - \varepsilon)c.$$

For $c > 1$ and ε sufficiently large, this quantity is strictly smaller than 1, so the remaining graph lies in the subcritical regime.

From the results established earlier for the subcritical case, it follows that all connected components in the remaining graph have size at most logarithmic in n with high probability. In particular, no component of linear size can exist in the remainder. This contradicts the assumption that C_2 has linear size.

We conclude that with high probability there can be at most one connected component of order $\Theta(n)$. Together with the existence of such a component in the supercritical regime, this proves that the Erdős–Rényi random graph contains a *unique* giant component.

4 Simulation of the Phase Transition

4.1 Description of the Simulation Setup

We are now going to numerically verify our results. For this we will use a Monte-Carlo simulation.

We fix the graph size at $n = 10^4$ vertices to minimize finite-size effects and the average degree parameter c will vary across the interval $[0, 2]$.

For each specific value of c , we perform $K = 100$ independent trials. In every trial, the simulation executes the following steps:

- (a) **Graph Generation:** A random graph is constructed where each of the $\binom{n}{2}$ possible edges is included independently with probability $p = c/n$.
- (b) **Measurement:** The connected components are identified using a Breadth-First Search, and the size of the largest component $|C_{\max}|$ is recorded.

Finally, we average the results over all K trials to estimate the expected size of the largest component. We then divide this value by n to determine the fraction of the network occupied by the giant component.

The code used for this visualization is available here. [3]

4.2 Visualization of the Subcritical Regime

In order to illustrate the structural changes induced by an increase in the average degree, we visualize typical realizations of the Erdős–Rényi random graph for several values of the expected degree c .

For visualization purposes, the nodes are ordered from left to right according to decreasing degree. This degree-ordered representation highlights the emergence of hubs, trees, and isolated vertices.

To ensure readability of edges and connectivity patterns, the visualized graphs contain fewer vertices ($n = 500$) than those used in the Monte Carlo simulations. Increasing the number of nodes further would not yield a more informative visualization due to excessive visual noise.

Expected degree $c = 0.3$. At this low expected degree, the graph is highly sparse. Even nodes with relatively high degree remain weakly connected, and the majority of vertices appear as isolated points. No large connected structures are present.

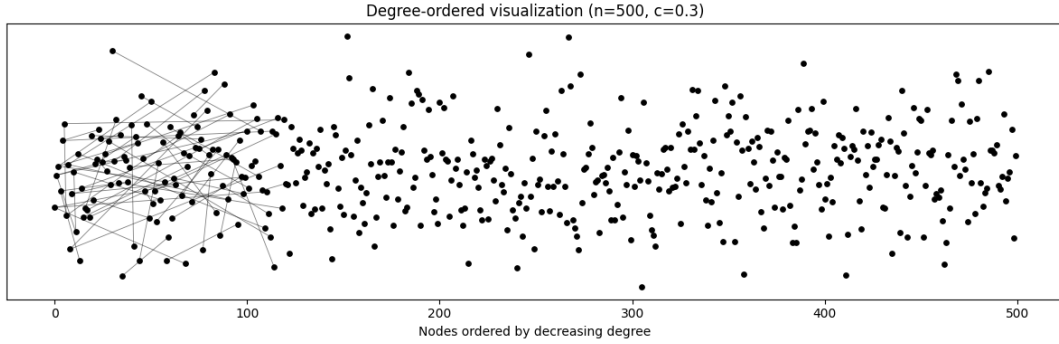


Figure 2: Degree-ordered visualization for $c = 0.3$.

Expected degree $c = 0.67$. As the expected degree increases, sparsity is reduced and small connected components begin to form. Nevertheless, most vertices are still disconnected, and no dominant component has yet emerged.

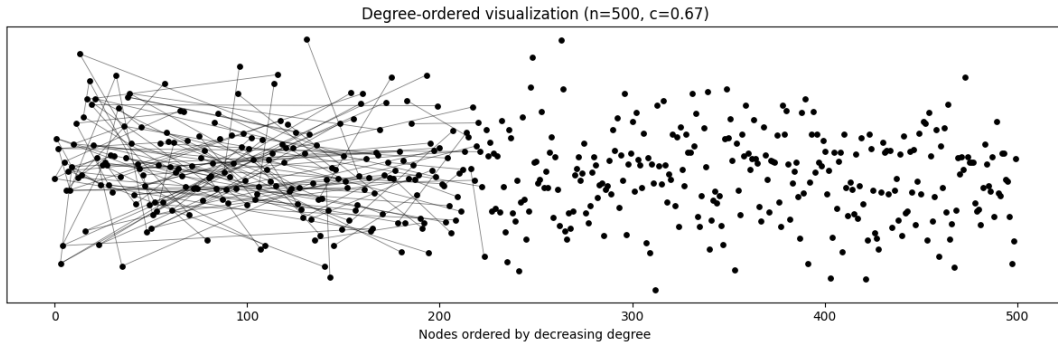


Figure 3: Degree-ordered visualization for $c = 0.67$.

Expected degree $c = 0.9$. Close to the critical regime, connectivity increases noticeably. The connected components become denser, and a larger fraction of vertices is absorbed into nontrivial structures, although the graph is still far from fully connected.

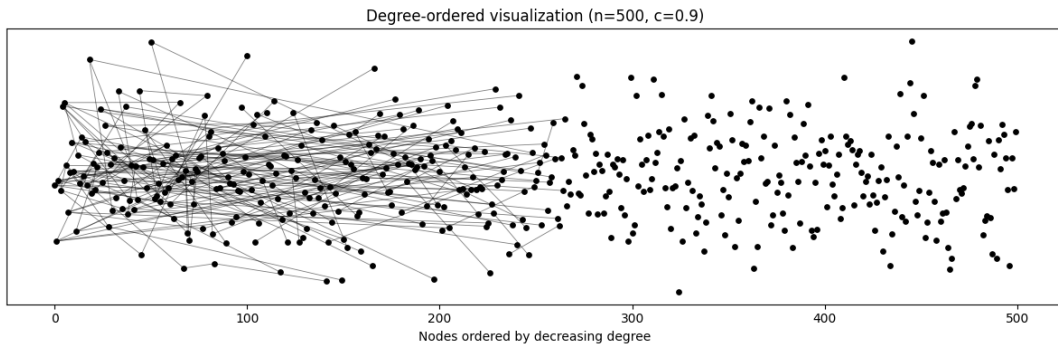


Figure 4: Degree-ordered visualization for $c = 0.9$.

Critical regime ($c \approx 1$). At the critical threshold, a pronounced change in structure becomes visible. Despite only a small increase in the expected degree, a large connected component emerges, signaling the onset of the phase transition predicted by theory.

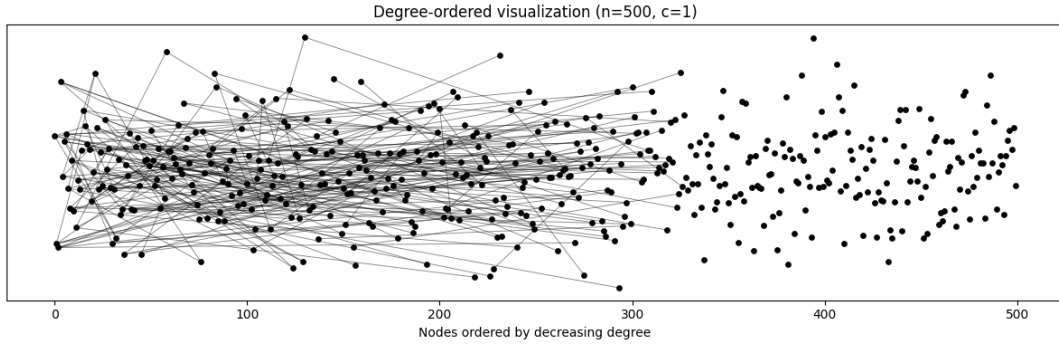


Figure 5: Degree-ordered visualization for $c = 1$.

4.3 Visualization of the Supercritical Regime

We have noticed an interesting behavior so far. However, it only gets better. We can notice a drastic surge in connectivity.

Expected degree $c = 1.2$. Slightly above the critical threshold, a dominant connected structure is already clearly visible. Compared to the critical regime, a substantially larger fraction of vertices is now part of a single connected component. Nevertheless, a noticeable number of isolated vertices and small components still persist on the right-hand side of the visualization. This reflects the early supercritical phase, where the giant component has emerged but has not yet absorbed the majority of the graph.

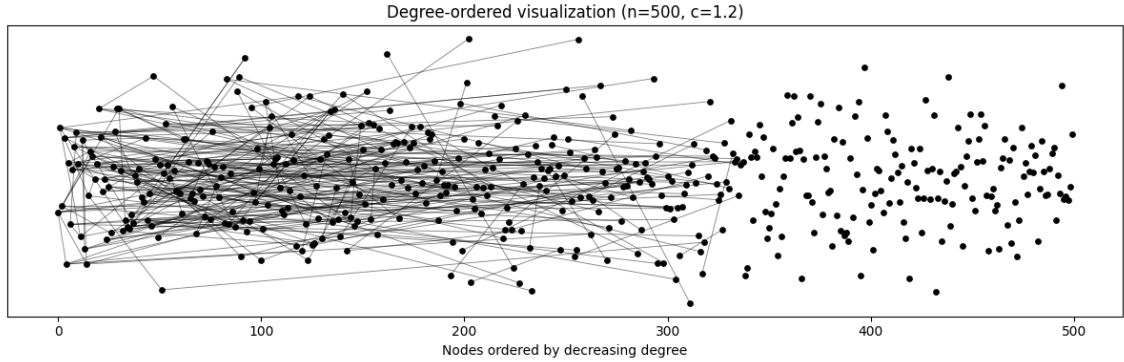


Figure 6: Degree-ordered visualization for $c = 1.2$.

Expected degree $c = 1.5$. As the expected degree increases further, a pronounced *boom* in connectivity becomes apparent. The giant component now spans a large portion of the graph and dominates the left side of the degree-ordered layout. Only a relatively small number of vertices remain isolated or belong to small components. This rapid expansion of the giant component illustrates the strongly supercritical nature of the graph.

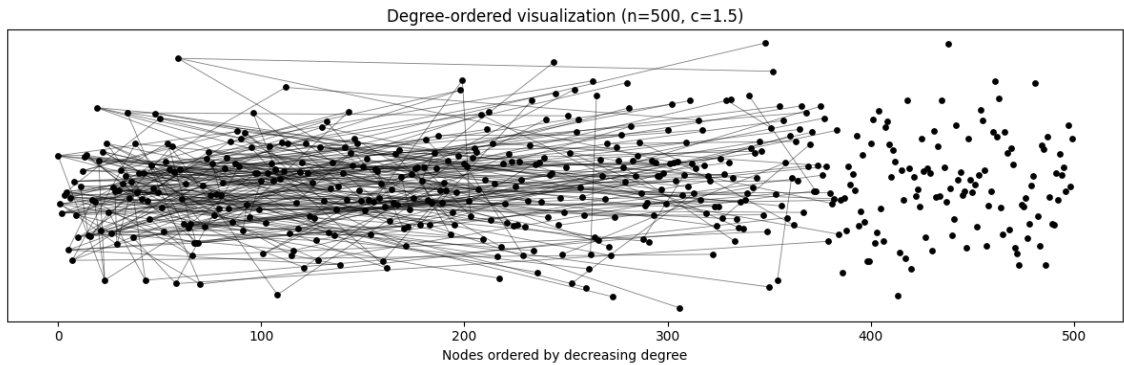


Figure 7: Degree-ordered visualization for $c = 1.5$.

Expected degree $c = 2.0$. Deep in the supercritical regime, the graph is overwhelmingly dominated by a single connected component. Almost all vertices are now part of the giant component, and isolated nodes become rare. The visualization highlights the saturation of connectivity: further increases in the expected degree primarily densify the existing giant component rather than creating new large-scale structural changes.

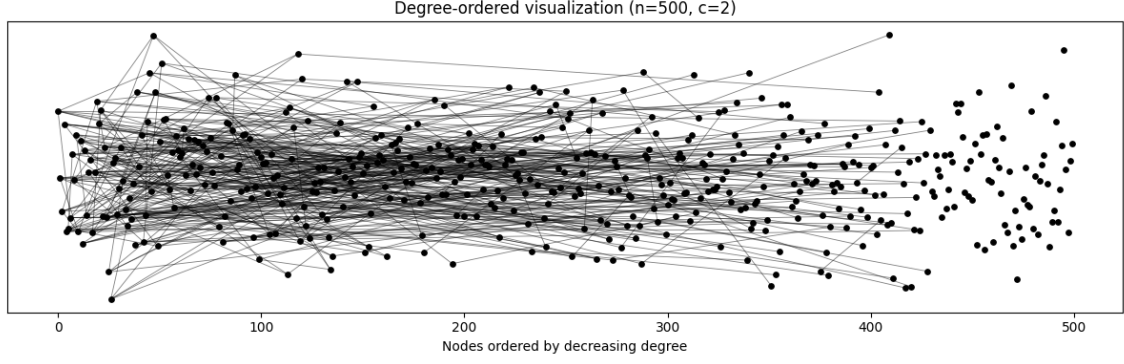


Figure 8: Degree-ordered visualization for $c = 2.0$.

4.4 Interpretation of Simulation Results

The Monte Carlo simulation [3] confirms the theoretical results derived in the previous sections and provides a clear numerical illustration of the phase transition in Erdős–Rényi random graphs.

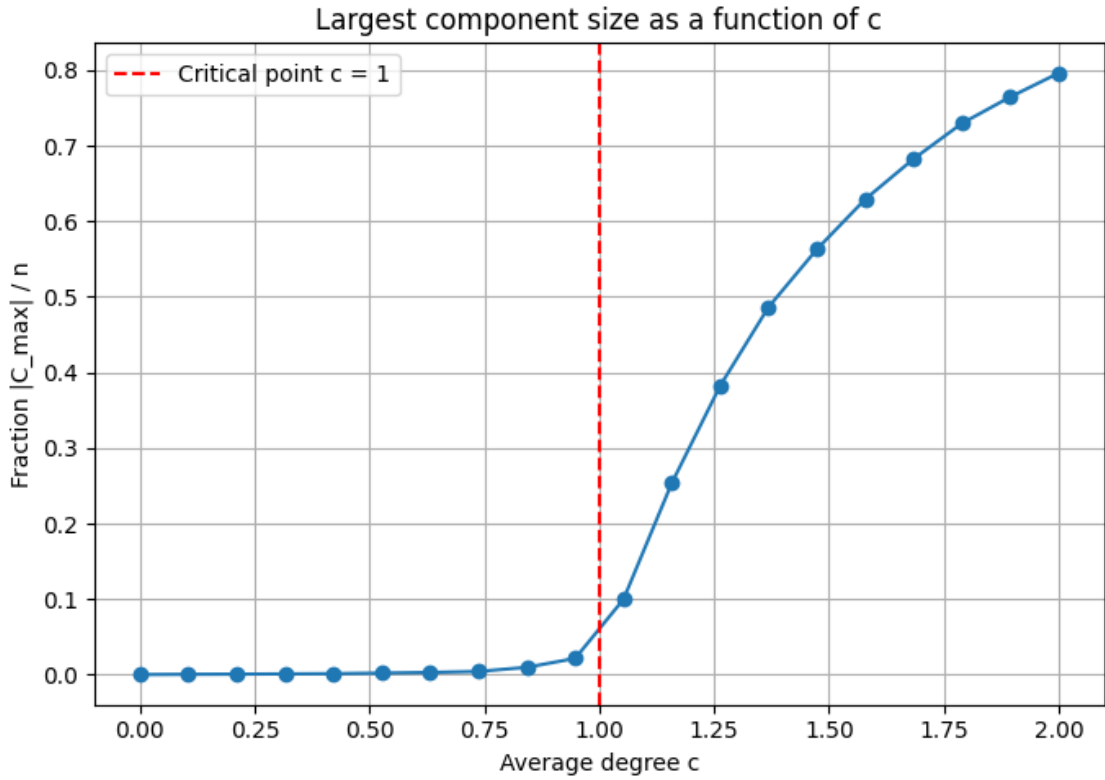


Figure 9: Estimated fraction of vertices belonging to the largest connected component, $\mathbb{E}[|C_{\max}|]/n$, as a function of the expected degree c .

Figure X shows the estimated fraction of vertices contained in the largest connected component as a function of the expected degree. For small values of c , the size of the largest component remains negligible. In particular, for $c \leq 0.75$, the simulation yields values of $\mathbb{E}[|C_{\max}|]/n$ close to zero, indicating that the graph consists almost entirely of small components and isolated vertices.

Expected degree c	$\mathbb{E}[C_{\max}]/n$
0.00	0.0001
0.11	0.0005
0.21	0.0007
0.32	0.0010
0.42	0.0013
0.53	0.0021
0.63	0.0029
0.74	0.0043
0.84	0.0097
0.95	0.0215
1.05	0.0997
1.16	0.2536
1.26	0.3823
1.37	0.4864
1.47	0.5630
1.58	0.6292
1.68	0.6826
1.79	0.7296
1.89	0.7646
2.00	0.7961

Table 1: Monte Carlo estimates of the fraction of vertices in the largest connected component for different values of the expected degree c .

As the expected degree approaches the critical threshold $c = 1$, a noticeable change in behavior occurs. While growth remains modest up to $c \approx 0.9$, a rapid increase in the size of the largest component is observed shortly thereafter. Between $c = 1$ and $c \approx 1.5$, the fraction of vertices belonging to the giant component grows very quickly, reflecting the onset of the supercritical regime and the emergence of a macroscopic connected structure.

For larger values of c , the growth of the giant component gradually slows down. This saturation effect arises because the graph is already highly connected, and further increases in the expected degree primarily densify the existing giant component rather than significantly increasing its relative size. Overall, the simulation results are in excellent agreement with the theoretical prediction of a phase transition at $c = 1$.

References

- [1] Alan Frieze and Michał Karoński. *Introduction to Random Graphs*. Cambridge University Press, 2015.
- [2] Remco van der Hofstad. *Random Graphs and Complex Networks, Volume I*. Cambridge University Press, 2017.
- [3] Andrei Ciuta. *Random Graphs – Math, Simulation, and Usage*. GitHub repository, 2025. <https://github.com/AndreiCiuta04/Random-Graphs---Math-simulation-usage>.